

CYBER SECURITY COURSE

Netcat Persistence and Remote Shell Analysis

Professional Case Report 17

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Case ID	CSC-2026-17
Discipline	Adversary simulation
Overall risk	Critical
Evidence status	Laboratory methodology documented

Purpose

Document how simple network utilities can create persistent access and identify host and network indicators for detection.

Scope note

This report documents work performed in an authorized training environment. Confirmed findings are separated from methodology and expected laboratory behavior.

Executive Summary

Netcat was used to demonstrate remote command-shell access, persistence as a service and process execution through WMI. The exercise showed that a small utility can produce critical impact when paired with service or WMI execution. Detection must correlate process ancestry, service creation and network behavior.

Case Profile

Scenario	Netcat was used to demonstrate remote command-shell access, persistence as a service and process execution through WMI.
Environment	Windows target and Kali Linux listener in an isolated network.
Tools	Netcat, WMIC, Windows services, netstat, Process monitoring

Methodology

I used the following sequence to keep the analysis repeatable and to separate observation from interpretation.

1. Established a temporary Netcat command shell.
2. Observed listening and established connections with netstat.
3. Tested persistence through a Windows service and removed the service after validation.
4. Compared direct and WMI-launched process behavior.
5. Documented cleanup of listeners and binaries.

Technical Findings

ID	Finding	Risk	Evidence and analysis
F01	Unauthenticated command shell	Critical	A listener bound to a command interpreter provides direct remote execution.
F02	Service-based persistence	Critical	A malicious service can restart remote-access tooling after reboot.
F03	WMI process launch	High	Remote or local WMI execution can reduce obvious console interaction while leaving process and event evidence.

Representative Commands and Actions

The following commands or analysis actions represent the core of the practical workflow.

```
netstat -nao | find ':4444'
wmic process call create <command>
sc delete ncservice2
```

Risk Assessment

Risk	Level	Operational impact
Persistent remote access	Critical	Attacker retains command execution
Weak attribution	High	Generic tools can blend with legitimate administration

Recommendations

1. Block unauthorized Netcat binaries through application control.
2. Alert on services launching command shells or tools from writable paths.
3. Monitor WMI process creation and port 4444 activity.
4. Verify service removal and close all lab listeners after testing.

Knowledge Demonstrated

- Remote-shell analysis
- Service persistence
- WMI execution
- Cleanup verification

Conclusion

The exercise showed that a small utility can produce critical impact when paired with service or WMI execution. Detection must correlate process ancestry, service creation and network behavior.

Hands On Lab Practical 17

Completed as part of the authorized cyber security course practical work.